

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: AZD2014 and Fulvestrant in Patients With ER+ Advanced Metastatic Breast Cancer **Official Title:** A Phase I, Open-label, Multicentre, Study to Assess the Safety, Tolerability, Pharmacokinetics and Preliminary Efficacy of AZD2014 Administered Orally in Combination With Intramuscular (IM) Fulvestrant to Patients With Estrogen Receptor Positive (ER+) Advanced, Metastatic Breast Cancer **Short Title/Acronym:** AZD2014 + Fulvestrant in ER+ MBC **Protocol Number:** D2270C00005 **NCT Number:** NCT01597388 **Posting ID:** 132817830602879682 **Sponsor/Company Name:** AstraZeneca **Investigational Product:** AZD2014 (vistusertib) and Fulvestrant **Phase of Development:** Phase I **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of AZD2014 when given in combination with fulvestrant, using continuous and intermittent dosing schedules, in patients with ER+ advanced metastatic breast cancer. **Secondary Objectives:** To determine the steady-state pharmacokinetic (PK) profile of AZD2014 in combination with fulvestrant, and to evaluate preliminary efficacy and anti-tumour activity, including Clinical Benefit Rate (CBR) and Overall Response Rate (ORR).

2.2 Background & Rationale Preclinical and clinical data suggest that patients with ER+ breast cancer lose sensitivity to hormonal therapy and increase dependency on the mTOR pathway. AZD2014 (vistusertib) is a potent, selective dual mTORC1/2 inhibitor. It offers potential additional benefits versus other mTOR inhibitors (like everolimus, which only inhibits mTORC1) by blocking mTOR complex 1 and 2, addressing a known resistance mechanism. This study aims to evaluate if the addition of AZD2014 can re-sensitize tumors to endocrine therapy (fulvestrant).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose escalation/expansion) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target Enrollment: 99 evaluable patients **Number of Sites:** Multiple sites in the United States

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Provision of signed and dated written informed consent prior to any study specific procedures, sampling analysis. 2. Aged at least 18 years. 3. Histological or cytological confirmation of an ER+ advanced metastatic breast cancer tumour that is eligible for treatment with fulvestrant. 4. At least one lesion (measurable and/or non-measurable) that can be accurately assessed at baseline by computerised tomography (CT), magnetic resonance imaging (MRI) or plain X-ray and is suitable for repeated assessment. 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 6. Patients must have evidence of non-child-bearing potential.

4.2 Exclusion Criteria 1. Prior chemotherapy, biological therapy, radiation therapy, androgens, thalidomide, immunotherapy, other anticancer agents, and any investigational agents within 14 days of starting study treatment (not including palliative radiotherapy at focal sites). 2. Major surgery within 4 weeks prior to entry to the study (excluding placement of vascular access), or minor surgery within 2 weeks of entry into the study. 3. Patients with severe cardiac condition of ischemia, impaired ventricular function and arrhythmias, evidence of severe or uncontrolled systemic or current unstable or uncompensated respiratory or cardiac conditions. 4. Patients with diabetes type 1 or uncontrolled type II (HbA1c > 8% assessed locally). 5. Pregnancy or breastfeeding (inferred from non-child-bearing potential requirement). 6. Unresolved toxicities from prior therapies (standard Phase I oncology criteria). 7. Concomitant use of prohibited medications, including strong CYP3A4 inhibitors/inducers.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **AZD2014:** Administered orally on either a continuous (Con) schedule (e.g., 50 mg twice daily [BD]) or an intermittent (Int) schedule (e.g., 125 mg BD for 2 out of 7 days). * **Fulvestrant (Trade Name: Faslodex | ATC Code: L02BA03):** 500 mg administered on Day 1 (with a Day 14 loading dose) of each 28-day cycle. **Route of Administration:** AZD2014 (Oral); Fulvestrant (Intramuscular injection). **Duration of Treatment:** 28-day cycles, continuing until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Prophylactic antiemetics to manage treatment-related nausea. **Prohibited:** Strong CYP3A4 inhibitors or inducers, and concurrent investigational anticancer agents.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Baseline Imaging (CT/MRI) | | **Baseline** | Day 1 | First Dose of AZD2014 + Fulvestrant 500mg, PK sampling | | **Interim** | Every 28-day cycle | Safety/Toxicity Evaluation, DLT Assessment, PK/PD monitoring | | **Assessment** | Week 16 & 24 | Tumor Assessment (Percentage Change in Target Lesion Size), CBR calculation | | **End of Study** | Until Progression | Final vital signs, exit interview, and follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- 1. Adverse Events:** Assessment of overall safety and tolerability via clinical observation, laboratory tests, and standard AE reporting graded by severity.
- 2. Adverse Events Leading to Dose Reduction of AZD2014:** Monitoring of Dose-Limiting Toxicities (DLTs) and tolerability.
- 3. Clinically Important Changes in Haematology Parameters:** Evaluation of laboratory safety signals.

7.2 Secondary & Exploratory Outcomes

- 1. AZD2014 Peak Plasma Concentration (C_{max}):** Evaluated following a single dose, fasted, no fulvestrant.
- 2. Time to AZD2014 Peak Plasma Concentration (T_{max}):** Evaluated following a single dose, fasted, no fulvestrant.
- 3. Area Under the Plasma Concentration-time Curve (AUC 0-12):** From 0 to 12 hours for AZD2014 following a single dose, fasted, no fulvestrant.
- 4. Clinical Benefit Rate (CBR):** Percentage of patients with confirmed CR, PR, or SD for ≥ 24 weeks.
- 5. Overall Response Rate (ORR):** Proportion of patients achieving CR or PR.
- 6. Target Lesion (TL) Size:** Percentage change from baseline at 16 weeks.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring throughout the study. Common AEs recorded included nausea, fatigue, stomatitis, rash, diarrhea, and hyperglycemia.

Serious Adverse Events (SAEs): Dose-limiting toxicities (DLTs) on the continuous schedule included rash/stomatitis, hyperglycemia, and fatigue. Discontinuation rates due to AEs were closely monitored (31% on Con; 3% on Int).

Data Monitoring Committee (DMC): Supervised by internal safety review committees typical of Phase I dose-escalation trials.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients receiving at least one dose) and Efficacy Evaluable Set. **Sample Size Justification:** Target enrollment of 99 was deemed appropriate to evaluate the maximum tolerated dose (MTD), recommended Phase 2 dose (RP2D), and preliminary efficacy signals in heavily pre-treated cohorts (median of 6 prior treatments). **Handling of Missing Data:** Standard Phase I oncology protocols for missing data; early withdrawals replaced if DLT evaluability was not met.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of trial sites to ensure adherence to continuous/intermittent dosing schedules. **Audit Trail:** Standard clinical database locking and verification mechanisms.

11. Study Identifiers & Governance

Primary ID: D2270C00005 **Registry Number:** NCT01597388 **Posting ID:** 132817830602879682 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** AZD2014 and Fulvestrant in Patients With ER+ Advanced Metastatic Breast Cancer

12. Clinical Framework (Breast Cancer Specific)

Disease Area: Advanced Metastatic Breast Cancer **Disease Name:** Breast Cancer **Preferred Name:** Malignant neoplasm of breast **Preferred UMLS Name:** Estrogen Receptor Positive Metastatic Breast Cancer **MeSH Code:** D001943 **SNOMED ID:** 254837009 **Diagnosis Threshold:** Histologically/cytologically confirmed ER+ and clinically eligible for fulvestrant therapy **Medical Methodology:** Endocrine therapy combined with targeted dual mTORC1/2 inhibition **Optimization Target:** Re-sensitization of tumor pathways to endocrine therapy

13. Study Procedures & Methodology

Management Pathway: Targeted oncology pathway for advanced MBC progression **Education Protocol (HCP):** Investigator training on recognizing and managing specific mTOR-inhibitor toxicities (e.g., stomatitis, hyperglycemia) **Education Protocol (Patient):** Counseling on strict adherence to the Con vs. Int dosing schedules and prophylactic antiemetic use **Quality Audit**

Mechanism: Real-time toxicity tracking and PK blood sampling coordination

14. Observation & Follow-Up Schedule

Total Duration: Median follow-up through to disease progression or withdrawal **Onsite Visit Cadence:** Day 1 and 14 of early cycles, followed by Day 1 of subsequent 28-day cycles **Remote Monitoring Frequency:** As per standard AE follow-up. **Checklist Review Interval:** Disease assessment scans evaluated every 8 weeks

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					